

- c. The client encrypts the challenge-text using the configured WEP key and sends the encrypted message to the access point.
- d. The access point decrypts the response and verifies if the decrypted text matches the challenge-text. It authenticates the client if the match is found.

In spite of the mechanisms used, this algorithm has a number of vulnerabilities and can be easily cracked. Various techniques, based on brute force attacks and analysis of the IVs (initialisation vectors), were discovered that led to deprecation of this algorithm.

Wi-Fi Protected Access (WPA)

WPA is a more secure algorithm that was developed in 2003 to address a few of the vulnerabilities that existed in WEP. WPA is mainly based on TKIP (temporal key integrity protocol), which uses a unique encryption key for each data packet sent over the network. The pre-shared key (PSK) used in TKIP is a 256-bit entity used for authentication. Figure 2 gives diagrammatic representation of WPA algorithm.

WPA is much more secure than WEP. This is because in the case of the latter, every data packet has the same key, which can be easy to find by capturing a sufficient number of packets. In WPA, it's difficult to get the key because every data packet has a unique key. But there are also a few loopholes that can be exploited. WPA can be compromised using Denial of Service attacks.

Wi-Fi Protected Access II (WPA2)

WPA2, also known as RSN (robust security network), is the most recent and highly secure algorithm, which enforces mandatory usage of the AES (advanced encryption standard). Another significant security enhancement has been the introduction of CCMP [counter mode with CBC (cipher block chaining) MAC (message authentication code) protocol]. CCMP uses AES instead of TKIP as the underlying encryption mechanism and, hence, prevents various attacks that were designed based on the RC4 cipher used in TKIP.

After some research, Eve concludes that WPA2 is secure enough to get rid of the kind of attacks Bob's office suffers from and, hence, configures the entire network over WPA2. Yet, even after such heightened countermeasures, Bob's network again gets compromised by some mischievous attackers. Alarmed at the situation, he again contacts Eve for help. During her research, she comes across one such tool that suits her needs. She advises Bob to use Aircrack-ng to internally spot the weak access points and enhance his network's security to avoid future attacks. Since Bob is unaware of the functionality of Aircrack-ng, Eve provides a brief overview.

Aircrack-ng stands for 'Aircrack new generation' and is an advanced network auditing software used for sniffing and cracking wireless networks. It is mainly used for testing the weaknesses of wireless networks by breaking into the

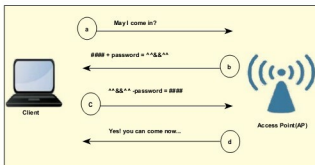


Figure 1: Shared key authentication mechanism

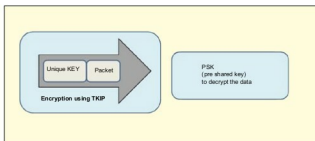


Figure 2: WPA technique

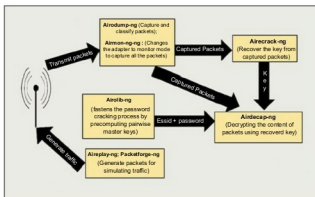


Figure 3: Block diagram of Aircrack-ng

network using the WEP and WPA-PSK keys recovered by decrypting the gathered encrypted packets. This tool can be used across Linux as well as Windows platforms, but has limited support in Windows.

The block diagram given in Figure 3 gives a brief description of the tool.

Bob is determined to patch up the weak links in his network; so he asks Eve to demonstrate how the tool is used to prevent various attacks. Eve tells him how to install the tool on the Ubuntu 14.04 platform and then gives the procedure to detect the vulnerable access points.

Installation

The package manager can be used to install the tool by using the following commands at the terminal:

```
$sudo apt-get install build-essential
```